

DIVIDE AND CONQUER

SAMPLE PROBLEMS



SCHOOL OF
INFORMATION TECHNOLOGY



FIND MAXIMUM ELEMENT

- **Problem:**

- [4, 9, 2, 7, 5, 8]

- **Divide:**

- Split into two halves:
- [4, 9, 2] [7, 5, 8]





FIND MAXIMUM ELEMENT

- **Problem:**

- [4, 9, 2, 7, 5, 8]

- **Conquer:**

- Find max of each half:

- **Left:**

- [4, 9, 2]
- **max = 9**

- **Right:**

- [7, 5, 8]
- **max = 8**





FIND MAXIMUM ELEMENT

- **Problem:**
 - [4, 9, 2, 7, 5, 8]
- **Combine:**
 - Compare results:
 - $\max(9, 8) = 9$





FIND MAXIMUM ELEMENT

- **Problem:**

- [4, 9, 2, 7, 5, 8]

- **Final Answer:**

- Maximum = 9





SUM OF ARRAY

- **Problem:**

- Find sum of: [2, 4, 6, 8]

- **Divide:**

- [2, 4] [6, 8]





SUM OF ARRAY

- **Problem:**

- Find sum of: [2, 4, 6, 8]

- **Conquer:**

- $2 + 4 = 6$
- $6 + 8 = 14$





SUM OF ARRAY

- **Problem:**

- Find sum of:[2, 4, 6, 8]

- **Combine:**

- $6 + 14 = 20$





SUM OF ARRAY

- **Problem:**

- Find sum of:[2, 4, 6, 8]

- **Final Answer:**

- Sum = 20





COUNT OCCURRENCES

- **Problem:**

- Count how many times 5 appears in:
- [5, 3, 5, 2, 5, 1]

- **Divide:**

- [5, 3, 5] [2, 5, 1]



COUNT OCCURRENCES

- **Problem:**

- Count how many times 5 appears in:
- [5, 3, 5, 2, 5, 1]

- **Conquer:**

- Left → 2 occurrences
- Right → 1 occurrence





COUNT OCCURRENCES

- **Problem:**

- Count how many times 5 appears in:
- [5, 3, 5, 2, 5, 1]

- **Combine:**

- $2 + 1 = 3$





COUNT OCCURRENCES

- **Problem:**

- Count how many times 5 appears in:
- [5, 3, 5, 2, 5, 1]

- **Final Answer:**

- 3 occurrences





POWER CALCULATION (FAST EXPONENTIATION)

- **Problem:**
 - Compute:
 - 2^8
- **Divide:**
 - $2^8 = (2^4)(2^4)$





POWER CALCULATION (FAST EXPONENTIATION)

- **Problem:**
 - Compute:
 - 2^8
- **Conquer:**
 - $2^4 = (2^2)(2^2)$
 - $2^2 = (2^1)(2^1)$





POWER CALCULATION (FAST EXPONENTIATION)

- **Problem:**

- Compute:
 - 2^8

- **Combine:**

- $2^2 = 4$
- $2^4 = 16$
- $2^8 = 256$





CLOSEST PAIR PROBLEM (DIVIDE AND CONQUER)

- **Points:**
 - $(1,1)$, $(3,4)$, $(6,1)$, $(2,2)$
- **Step 1: Sort Points**
 - Sort points by x-coordinate
 - Example:
 - $(1,1)$, $(2,2)$, $(3,4)$, $(6,1)$





CLOSEST PAIR PROBLEM (DIVIDE AND CONQUER)

- **Points:**

- $(1,1)$, $(3,4)$, $(6,1)$, $(2,2)$

- **Step 2: Divide**

- Split into two halves:
 - Left: $(1,1)$, $(2,2)$
 - Right: $(3,4)$, $(6,1)$



CLOSEST PAIR PROBLEM (DIVIDE AND CONQUER)

- **Points:**

- $(1,1), (3,4), (6,1), (2,2)$

- **Step 3: Conquer**

- Recursively compute:
- Closest in LEFT $\rightarrow$ distance = d_1
- Closest in RIGHT $\rightarrow$ distance = d_2

- **Left distance = $\sqrt{2} \approx 1.41$**

- **Right distance = $\sqrt{13}$**

- **So: $d = 1.41$**





CLOSEST PAIR PROBLEM (DIVIDE AND CONQUER)

- **Points:**

- $(1,1)$, $(3,4)$, $(6,1)$, $(2,2)$

- **Sorted:**

- $(1,1)$, $(2,2)$, $(3,4)$, $(6,1)$

- **Strip:**

- Points within distance d from middle:
 - $(2,2)$, $(3,4)$

- **Distance:** $\sqrt{5} \approx 2.24$





CLOSEST PAIR PROBLEM (DIVIDE AND CONQUER)

- **Points:**

- $(1,1), (3,4), (6,1), (2,2)$

- **Sorted:**

- $(1,1), (2,2), (3,4), (6,1)$

- **Final Answer:**

- Closest distance = 1.41





MAJORITY ELEMENT PROBLEM

- **Task:**
 - Find an element that appears more than $n/2$ times
- **Given:**
 - $A = [1, 2, 1, 1, 3, 1, 4, 1]$



MAJORITY ELEMENT PROBLEM

- **Task:**
 - Find an element that appears more than $n/2$ times
- **Divide:**
 - Left: [1, 2, 1, 1]
 - Right: [3, 1, 4, 1]
- **Given:**
 - $A = [1, 2, 1, 1, 3, 1, 4, 1]$



MAJORITY ELEMENT PROBLEM

- **Task:**

- Find an element that appears more than $n/2$ times

- **Given:**

- $A = [1, 2, 1, 1, 3, 1, 4, 1]$

- **Conquer:**

- LEFT:
 - $[1, 2, 1, 1] \rightarrow \text{majority} = 1$
- RIGHT:
 - $[3, 1, 4, 1] \rightarrow \text{no clear majority (returns candidate} \rightarrow 1 \text{ or none)}$



MAJORITY ELEMENT PROBLEM

- **Task:**

- Find an element that appears more than $n/2$ times

- **Given:**

- $A = [1, 2, 1, 1, 3, 1, 4, 1]$

- **Combine:**

- Compare candidates:
 - Left = 1
 - Right = 1 or none
- Count globally:
 - $\text{count}(1) = 5 > 4$

- **Final Answer:**

- Majority = 1





KEY TAKEAWAY

- **Divide and Conquer works best when:**
 - Problem can be divided into independent subproblems
 - Solutions can be combined efficiently
 - Recursion reduces complexity significantly



END



SCHOOL OF
INFORMATION TECHNOLOGY